

Mock Exam

Question 1

Not yet
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v1 (latest)

In standard, untyped Tree-based Genetic Programming (Koza-style), the property of **closure** is a strict requirement. Which of the following best describes what closure enforces?

- ☐ a. Every tree in the population must evaluate to a boolean value at its root node to allow selection to occur.
- ☐ b. The mutation operator must only replace a subtree with an identical syntactic structure to preserve tree depth.
- ☐ c. The terminal set must contain only constants, ensuring that execution terminates within a finite number of steps.
- ☐ d. Any function in the function set must be able to accept any value returned by any other function or terminal as an argument.

Question 2

Not yet
answered

v1 (latest)

In standard Particle Swarm Optimization, a particle updates its velocity based on three components: its current inertia, a cognitive component (p_{best}), and a social component (g_{best}). If you set both the cognitive acceleration coefficient (c_1) and the social acceleration coefficient (c_2) to exactly 0, how will the particles behave?

- ☐ a. The particles will immediately freeze in place at their current spatial coordinates.
- ☐ b. The particles will continue moving in straight lines along their current velocity vectors, completely ignoring their own past success and the success of the swarm.
- ☐ c. The particles will instantly collapse into the geometric center of the initial swarm.
- ☐ d. The particles will perform a pure random walk, as the velocity vector becomes entirely determined by stochastic noise.

Question 3

Not yet
answered

v1 (latest)

In **Linear Genetic Programming (LGP)**, programs are represented as sequences of instructions operating on a set of registers. What is a key structural characteristic of LGP that explicitly distinguishes it from tree-based GP variants regarding instruction execution?

- ☐ a. LGP variations must map every register back to a separate leaf node in a Context-Free Grammar prior to fitness evaluation.
- ☐ b. LGP contains structural introns—instructions whose outputs do not affect the final program output, meaning segments of the genotype can be entirely neutral.
- ☐ c. LGP completely eliminates the possibility of infinite loops because all programs must follow strict tree-depth limits.
- ☐ d. LGP requires all instructions to return boolean values to ensure proper evaluation execution.

Question 4

Not yet
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v1 (latest)

An ant at node i in an Ant Colony Optimization algorithm must choose the next edge to transition to. There are two available paths out of node i : path A and path B. The parameters are configured with $\alpha = 1$ (pheromone importance) and $\beta = 2$ (heuristic importance). The current matrix states are:

- Path A: Pheromone $\tau_A = 2$, Heuristic visibility $\eta_A = 3$
- Path B: Pheromone $\tau_B = 4$, Heuristic visibility $\eta_B = 1$

What is the exact probability P_A that the ant selects **Path A**?

- ☐ a. 0.818
- ☐ b. 0.500
- ☐ c. 0.600
- ☐ d. 0.182

Question 5

Not yet
answered

v1 (latest)

In Grammatical Evolution (GE), the genotype is a variable-length string of integers (codons), and the phenotype is a program generated via a Context-Free Grammar (CFG). During the mapping process, a specific non-terminal node has 3 production rules available in the grammar. If the current

codon value being read from the genotype is **26**, which production rule will be selected?

- ☐ a. The 3rd rule (Index 2)
- ☐ b. The 2nd rule (Index 1)
- ☐ c. The 1st rule (Index 0)
- ☐ d. The mapping fails, triggering structural wrapping immediately.

Question 6

Not yet
answered

v1 (latest)

In a competitive co-evolutionary system (e.g., a predator-prey or attacker-defender configuration), algorithms often face the challenge of **the Red Queen Effect**. Which of the following options precisely encapsulates why standard fitness metrics break down under this effect?

- ☐ a. The fitness landscape is dynamic and relative; as individuals in population A adapt to outperform population B, population B also shifts, meaning absolute generational fitness improvements can appear flat or non-existent.
- ☐ b. The population sizes of both species oscillate according to strict Lotka-Volterra differential equations, leading to mandatory premature convergence.
- ☐ c. Both populations quickly discover structural introns that block any variation operators from altering phenotypic outcomes.
- ☐ d. The selection pressure drops to zero because individual successes are evaluated entirely on independent, fixed objective test benches.

Question 7

Not yet
answered

v1 (latest)

A student is adapting a standard Genetic Algorithm to optimize a high-precision engineering problem involving continuous variables. They decide to switch from a classic binary string representation (using standard binary-to-gray code mapping) to a continuous real-valued vector representation. Assuming optimal variation operators are used for both, what is a fundamental structural advantage of the real-valued representation in this context?

- ☐ a. It prevents the phenomenon of genetic drift across the population.

- ☐ b. It eliminates the problem of 'Hamming cliffs', where adjacent phenotypic values require multiple simultaneous bit flips.
- ☐ c. It completely removes the need for mutation operators, relying solely on directional crossover.
- ☐ d. It guarantees that the schema theorem holds strictly, which is mathematically impossible with binary representations.

Question 8

Not yet
answered

v1 (latest)

What is a critical operational difference between a (μ, λ) Evolutionary Strategy and a $(\mu + \lambda)$ Evolutionary Strategy?

- ☐ a. In (μ, λ) , the parents are completely discarded, and the next generation's μ parents are selected *only* from the λ new offspring.
- ☐ b. (μ, λ) requires that $\mu > \lambda$, whereas $(\mu + \lambda)$ requires that $\lambda > \mu$.
- ☐ c. In (μ, λ) , the mutation rate is automatically adapted, whereas in $(\mu + \lambda)$ it remains static.
- ☐ d. (μ, λ) guarantees elitism, while $(\mu + \lambda)$ allows the average fitness of the population to decrease over generations.

Question 9

Not yet
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v1 (latest)

In Ant Colony Optimization, if the pheromone evaporation rate parameter (ρ) is set too **high** (i.e., approaching 1), what failure mode is most likely to occur in the swarm's behavior?

- ☐ a. The colony will fail to build up structural memory of good paths, causing the ants to behave almost randomly and lose exploitation capability.
- ☐ b. The swarm will experience stagnation, where all ants permanently follow the exact same sub-optimal trail.
- ☐ c. The heuristic information (η) will completely override the pheromone matrix, turning the algorithm into a deterministic greedy search.
- ☐ d. The convergence rate will accelerate exponentially, finding the global optimum within the first few iterations.

Question 10

Not yet
answered

v1 (latest)

Consider two parents in a real-valued GA with single-variable chromosome values $x_1 = 2.0$ and $x_2 = 6.0$. If we apply Blend Crossover (BLX- α) with $\alpha = 0.5$, which of the following ranges represents the complete interval from which an offspring's value can be uniformly sampled?

- ☐ a. [1.0, 7.0]
- ☐ b. [-2.0, 10.0]
- ☐ c. [2.0, 6.0]
- ☐ d. [0.0, 8.0]

Question 11

Not yet
answered

v1 (latest)

You run a Genetic Algorithm using **Tournament Selection**. Midway through the run, you observe that the algorithm is converging prematurely to local optima. You want to modify the tournament size k to *reduce* the selection pressure and encourage more exploration. What adjustment should you make?

- ☐ a. Increase k , because a larger tournament increases the probability that lower-fitness individuals are selected.
- ☐ b. Set $k = 1$, which maximizes selection pressure by enforcing strict elitism.
- ☐ c. Switch to a dynamic k that scales proportionally with the maximum fitness of the population.
- ☐ d. Decrease k , because a smaller tournament reduces the probability that the absolute best individuals dominate the selection pools.

Question 12

Not yet
answered

v1 (latest)

A particle in a 2D Particle Swarm Optimization space currently resides at position vector $X_t = [4.0, -2.0]$ with an instantaneous velocity vector $V_t = [1.0, 3.0]$. Through evaluation, the swarm determines the velocity update for the next time step to be $V_{t+1} = [-2.0, 1.5]$. What will the particle's updated position vector X_{t+1} be?

- ☐ a. [2.0, 1.0]

- ☐ b. [3.0, 4.5]
- ☐ c. [5.0, 1.0]
- ☐ d. [2.0, -0.5]

Question 13

Not yet
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v1 (latest)

Consider a Genetic Algorithm population containing four individuals (A, B, C, D) with the following fitness values (where fitness is being maximized): Fitness(A) = 10, Fitness(B) = 20, Fitness(C) = 5, Fitness(D) = 15.

If the algorithm applies standard **Roulette Wheel Selection** (Fitness-Proportionate Selection), what is the probability that individual B is selected exactly twice in two independent selection spins?

- ☐ a. 0.16 (or 16%)
- ☐ b. 0.04 (or 4%)
- ☐ c. 0.40 (or 40%)
- ☐ d. 0.25 (or 25%)